Information Visualization

Theme 4. Tasks and Evaluation

This lecture is inspired by the lectures of
Tamara Munzner and Camilla Forsell

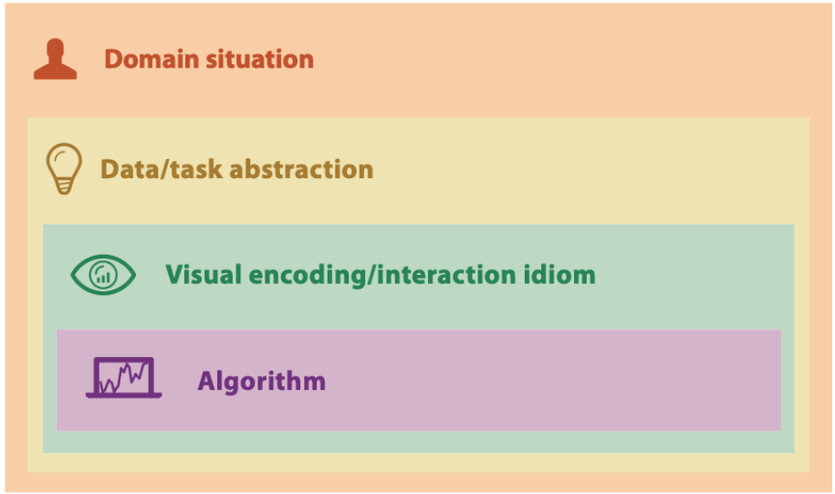
Agenda

- Task Abstraction (Ch 3 - Tamara's Book)
- Analysis: Nested Model (Ch 4 - Tamara's Book)
- Evaluation (User studies)
- Evaluation (Case studies)

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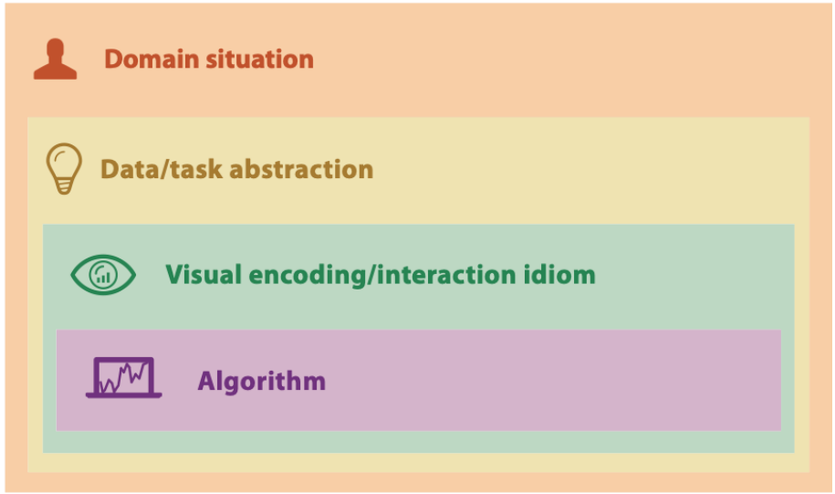
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From domain to abstraction



From domain to abstraction

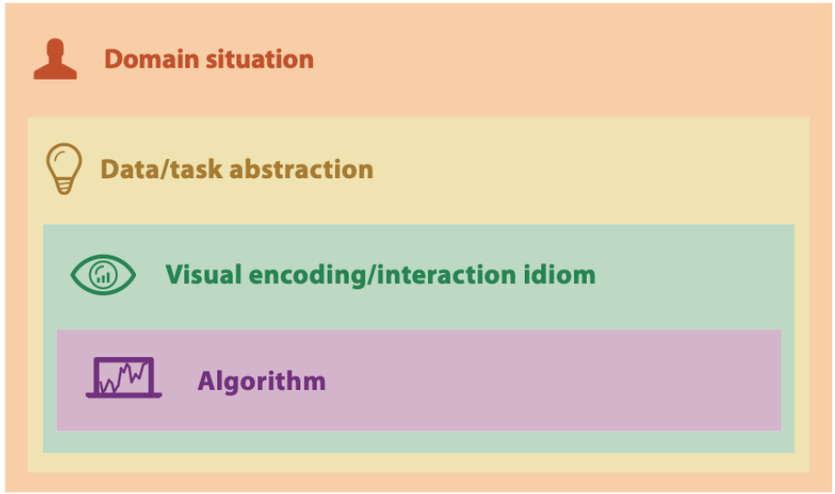
- domain characterization:
  details of application domain



domain

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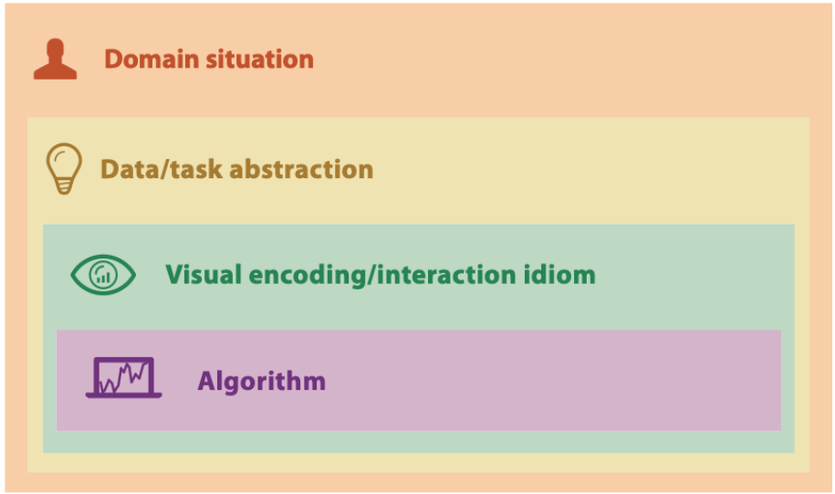
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  - group of users, target domain, their questions & data
    - varies wildly by domain
    - must be specific enough to get traction

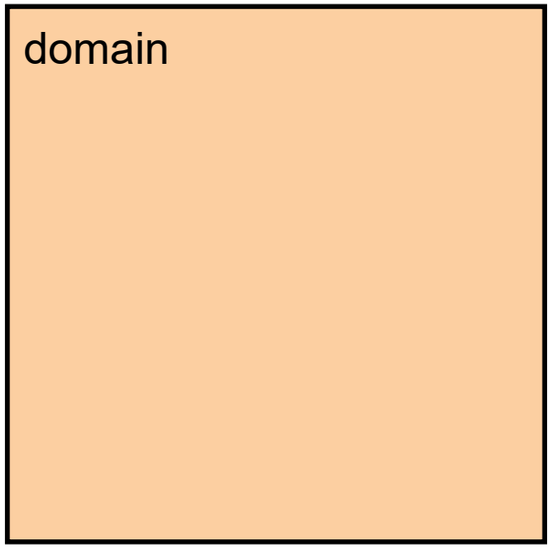


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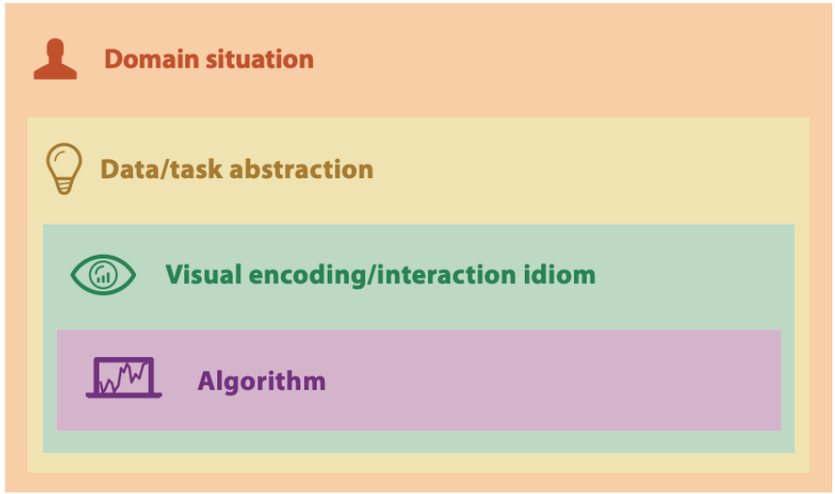
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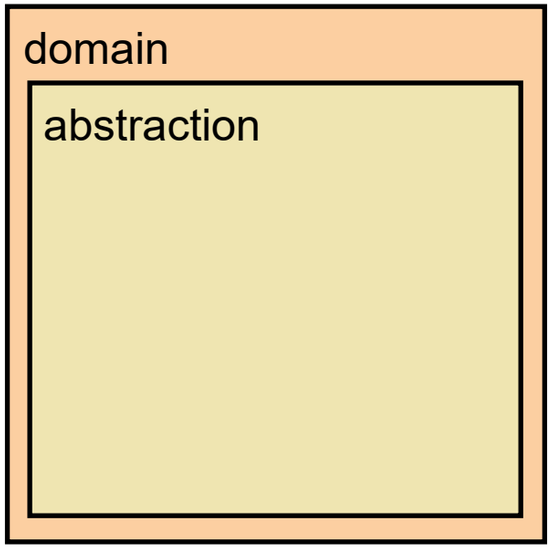




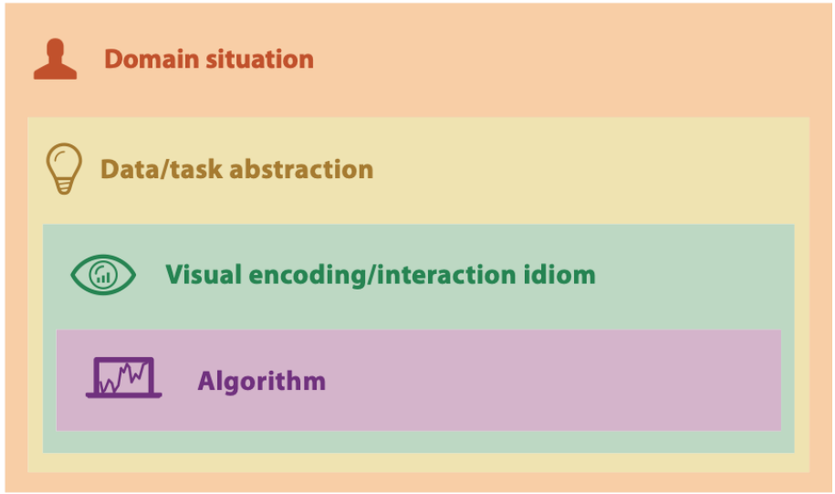
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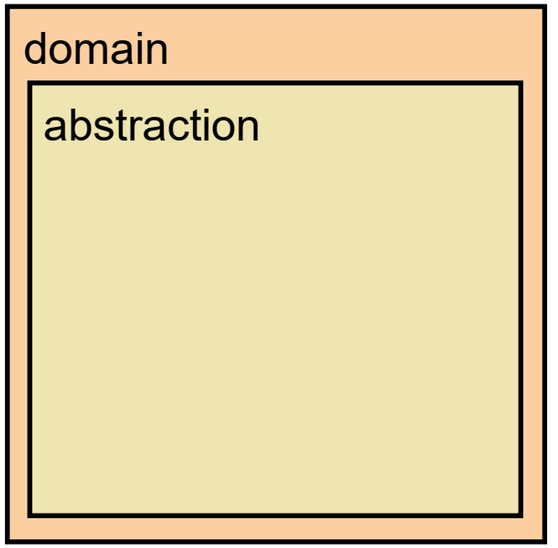




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  - group of users, target domain, their questions & data
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    - break down into simpler abstract tasks
- abstraction: data & task
  - map *what* and *why* into generalized terms
    - identify tasks that users wish to perform, or already do
    - find data types that will support those tasks



Task abstraction: Actions and targets

- very high-level pattern

- {action, target} pairs
  - *discover distribution*
  - *compare trends*
  - l*ocate outliers*
  - *browse topology*

Task abstraction: Actions and targets

- very high-level pattern
- actions
  - analyze
    - high-level choices
  - search
    - find a known/unknown item
  - query
    - find out about characteristics of item

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  - search
    - find a known/unknown item
  - query
    - find out about characteristics of item
- targets
  - what is being acted on

- {action, target} pairs
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Actions: Analyze

- consume
  - discover vs present
    - classic split
    - aka explore vs explain

➔ **Analyze**

➔ Consume

➔ *Discover*



➔ *Present*



Actions: Analyze

- consume
  - discover vs present
    - classic split
    - aka explore vs explain
- produce (data) by
  - annotate, record
  - derive
    - crucial design choice

➔ **Analyze**

➔ Consume

➔ *Discover*

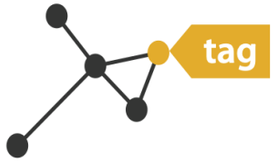


➔ *Present*



➔ Produce

➔ *Annotate*



➔ *Record*



➔ *Derive*



Actions: Search

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- what does user know?
  - target, location

Search

	Target known	Target unknown
Location known	 *Lookup*	 *Browse*
Location unknown	 *Locate*	 *Explore*

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  - ex: Scanning a country map to find where “Sundsvall” is

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Is this a **Locate** or **Lookup** task?

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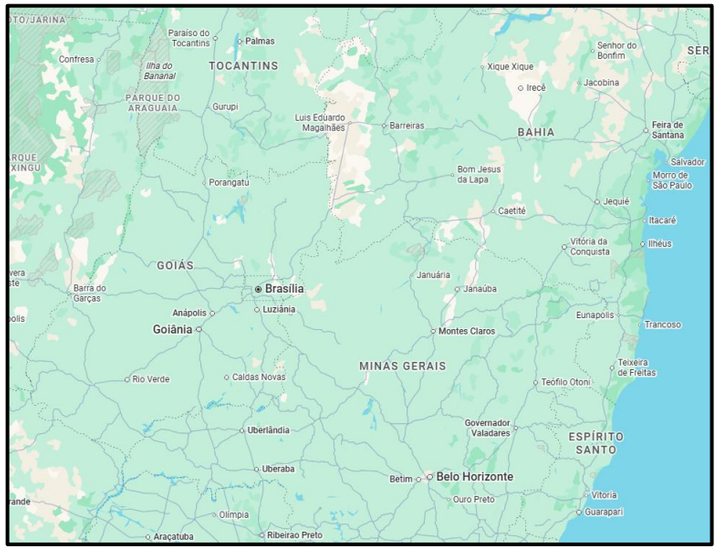
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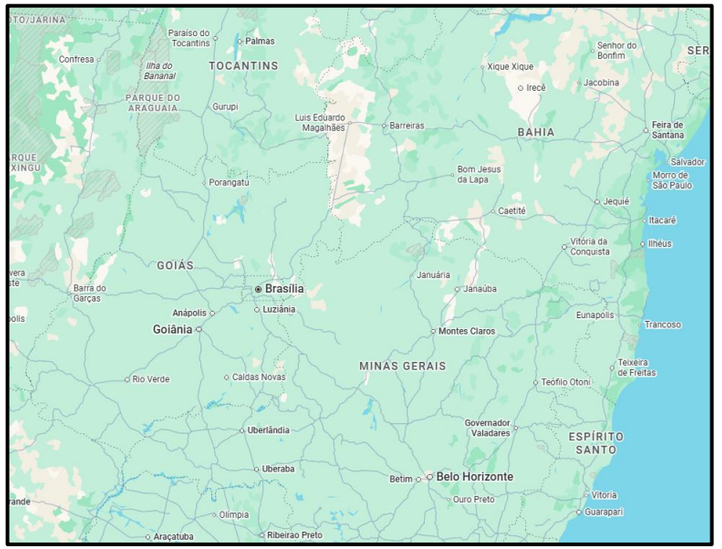
Depends on the target user

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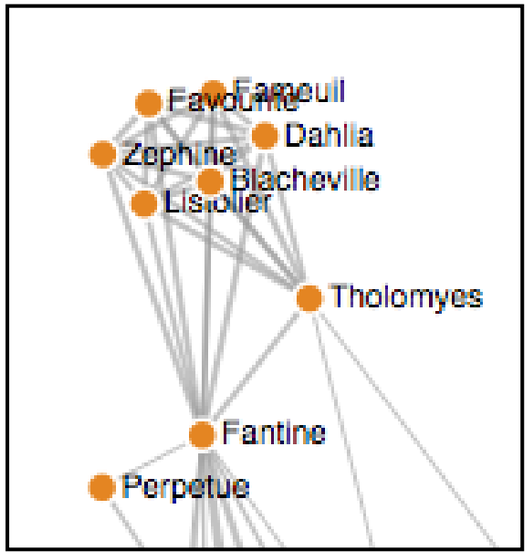
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  - ex: Scanning a country map to find where “Sundsvall” is
  - ex: node in network

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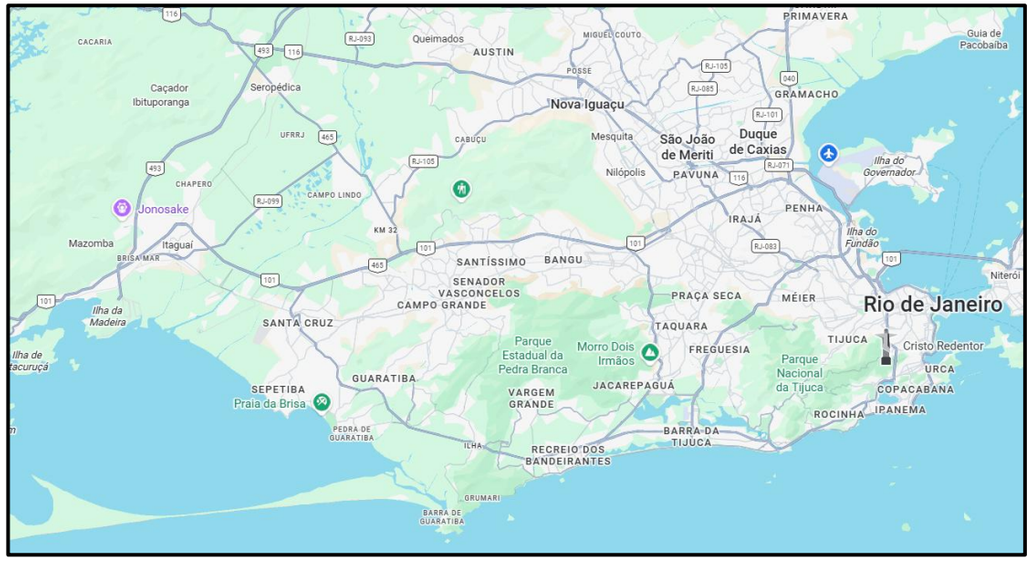
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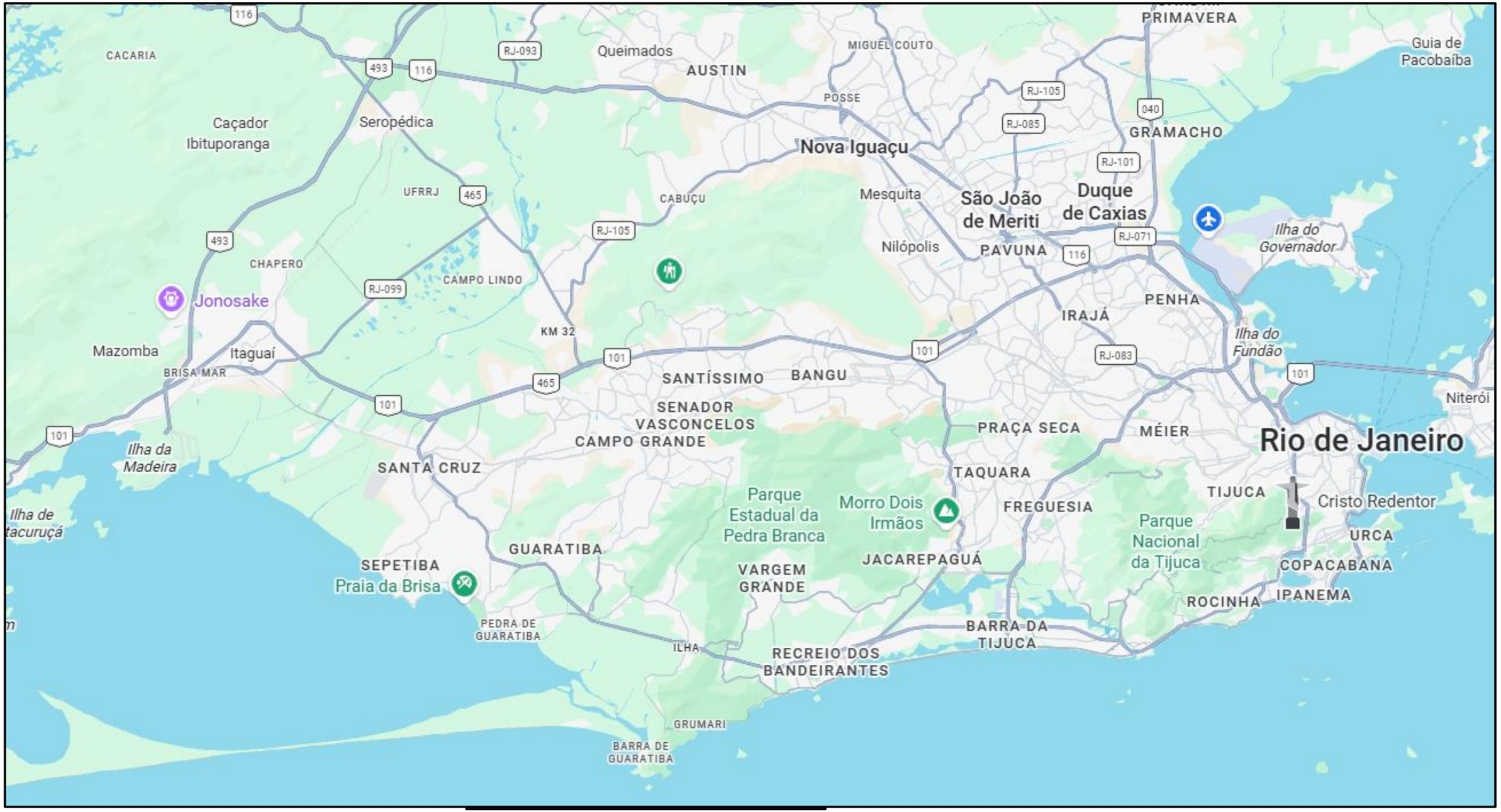
- browse
  - ex: Browsing a map to see all cities near a coastline
- explore
  - ex: Exploring a city map to discover “cool locations” to visit

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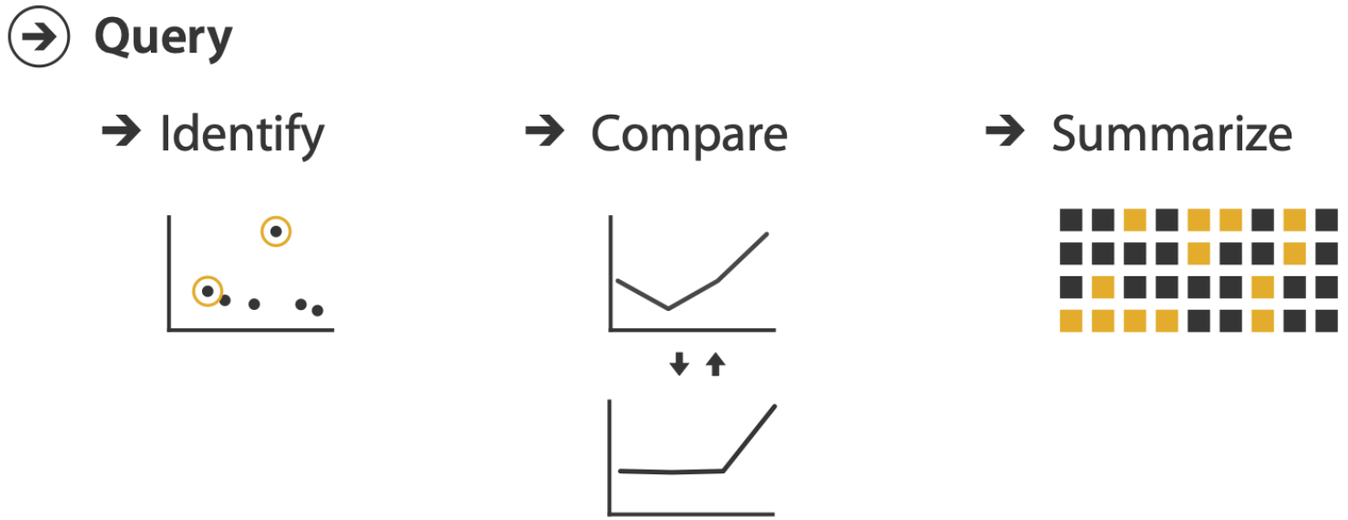





PRIMAVERA
Guia de Pacobaíba
CACARIA
Queimados
AUSTIN
MIGUEL COUTO
POSSE
Caçador
Ibituporanga
Seropédica
Nova Iguaçu
GRAMACHO
UFRRJ
CABUÇU
Mesquita
São João de Meriti
Duque de Caxias
Ilha do Governador
Nilópolis
PAVUNA
CHAPERO
CAMPO LINDO
Jonosake
PENHA
IRAJÁ
KM 32
Ilha do Fundão
Mazomba
Itaguaí
BRISA MAR
SANTÍSSIMO
BANGU
Niterói
SENADOR VASCONCELOS
CAMPO GRANDE
PRAÇA SECA
MÉIER
Rio de Janeiro
Ilha da Madeira
SANTA CRUZ
TAQUARA
TIJUCA
Parque Estadual da Pedra Branca
Morro Dois Irmãos
FREGUESIA
Cristo Redentor
Ilha de tacuruçá
Parque Nacional da Tijuca
URCA
GUARATIBA
JACAREPAGUÁ
COPACABANA
SEPETIBA
Praia da Brisa
VARGEM GRANDE
ROCINHA
IPANEMA
PEDRA DE GUARATIBA
BARRA DA TIJUCA
ILHA
RECREIO DOS BANDEIRANTES
GRUMARI
BARRA DE GUARATIBA

Actions: Query

- how much of the data matters?
  - one: identify
  - some: compare
  - all: summarize

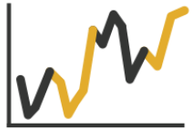


Task abstraction: Targets

- **All Data**
  - Trends
  - Outliers
  - Features





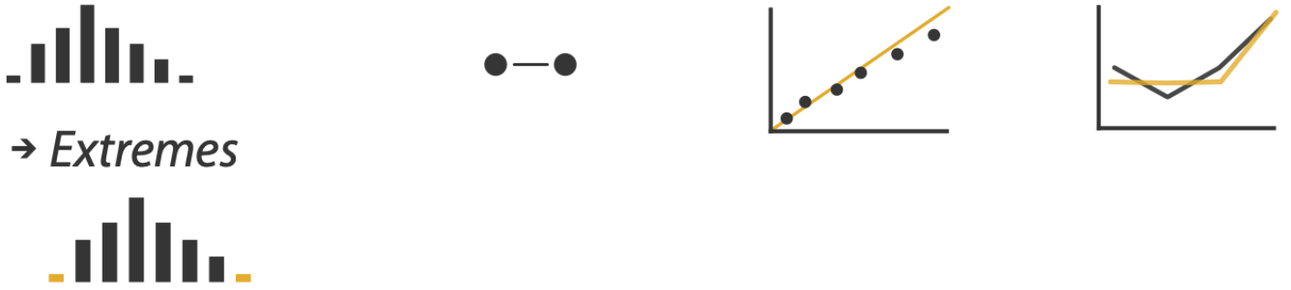


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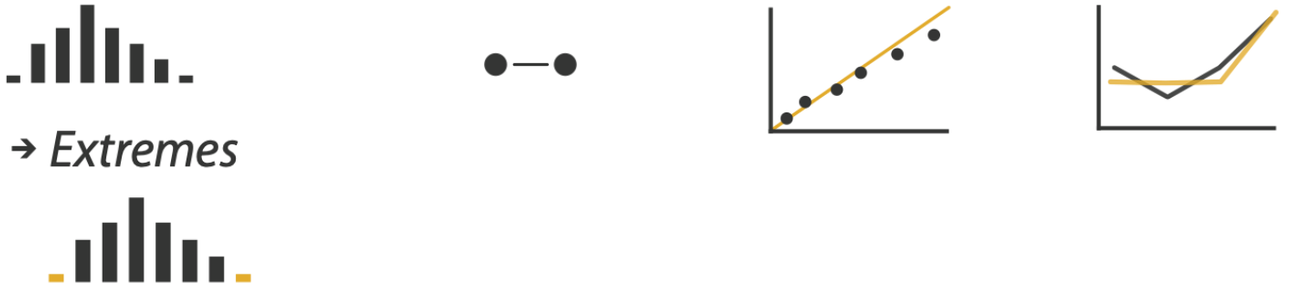


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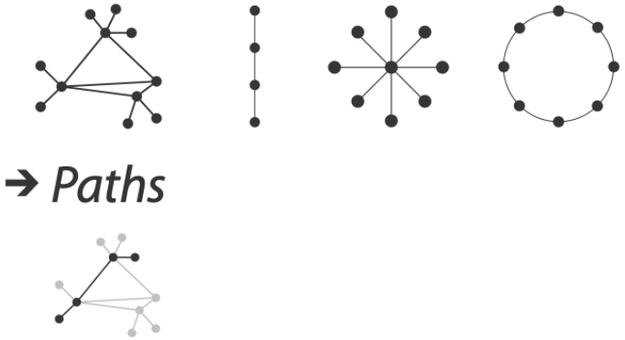
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- **Network Data**
  - Topology
  - *Paths*



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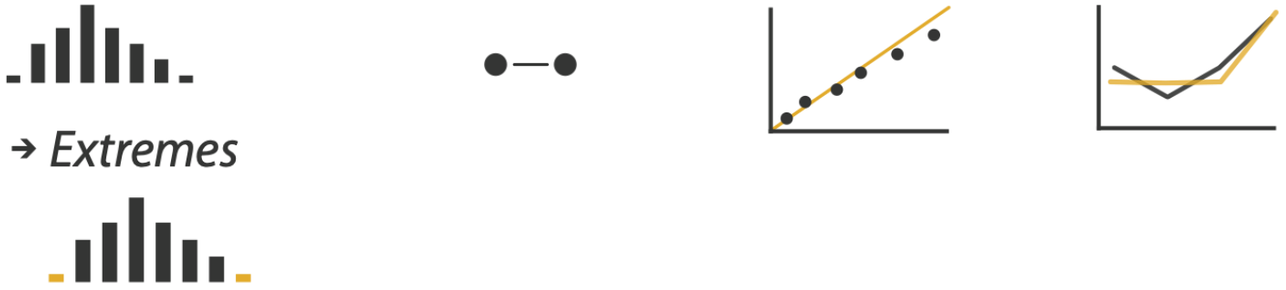
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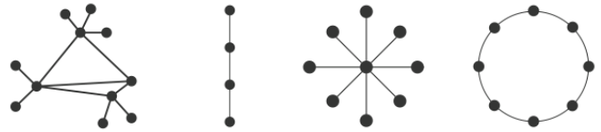
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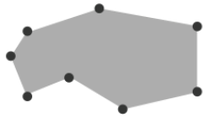


- *Paths*



Spatial Data

- Shape

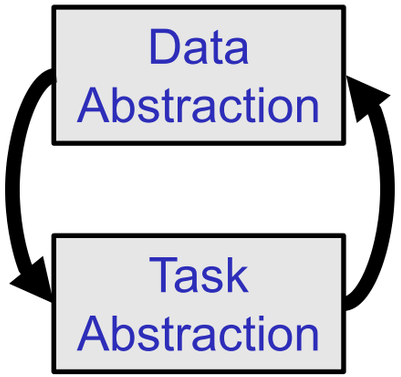


Abstraction

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- rule of thumb
  - systematically remove all domain jargon

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  - but sometimes you'll need more precision!
- rule of thumb
  - systematically remove all domain jargon
- interplay: task and data abstraction
  - need to use data abstraction within task abstraction
    - to specify your targets!
    - but task abstraction can lead you to transform the data
  - iterate back and forth
    - first pass data, first pass task, second pass data, ...



Example of Abstraction

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- Then you might realize your data needs to change (“We realize we need to aggregate by month”).
- Updating the task (“Now users can compare trends across seasons”).
- Then go back and refine both your data and task descriptions.

Example of Abstraction

- How to understand the {action, target} on tasks:

- We have customer transactions”
- Users want to find purchase trends” (task)
- “We realize we need to aggregate by month”
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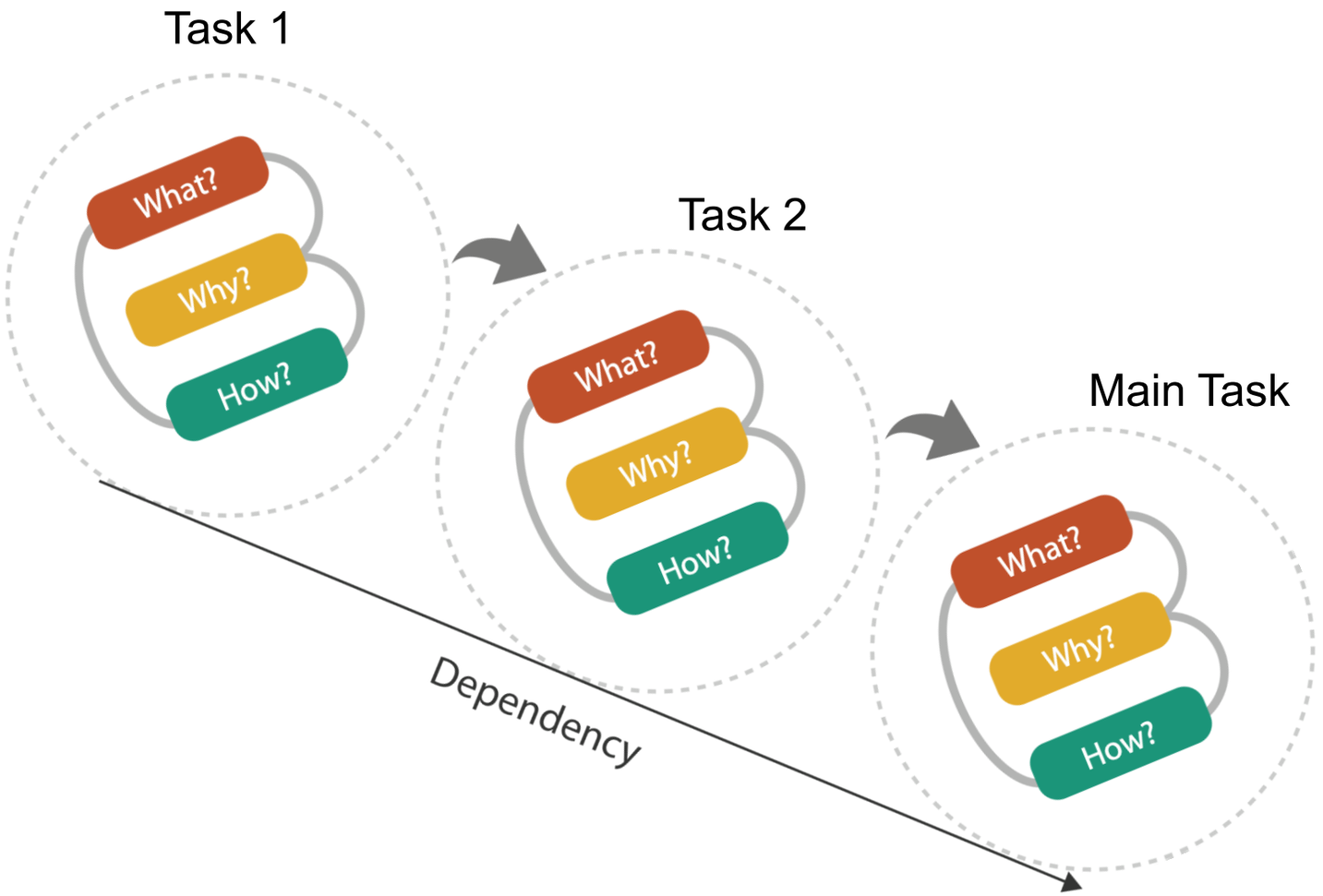
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Means and ends



Actions

Analyze

- Consume
  - *Discover*
  - *Present*



- Produce
  - *Annotate*
  - *Record*
  - *Derive*

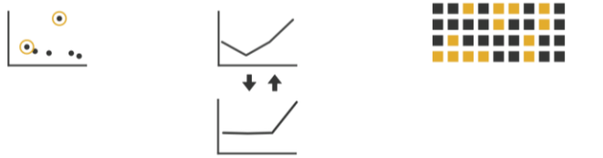


Search

	Target known	Target unknown
Location known	*Lookup*	*Browse*
Location unknown	*Locate*	*Explore*

Query

- Identify
- Compare
- Summarize



Targets

All Data

- Trends
- Outliers
- Features



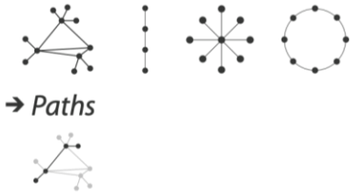
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Network Data

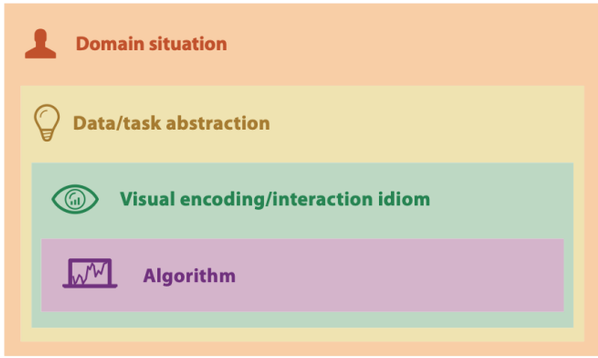
- Topology
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Spatial Data

- Shape





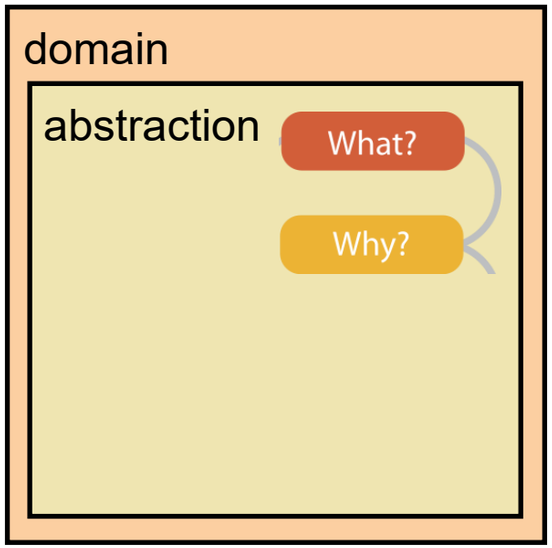
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Analysis framework: Four levels, three questions

- *domain situation*
  - who are the target users?

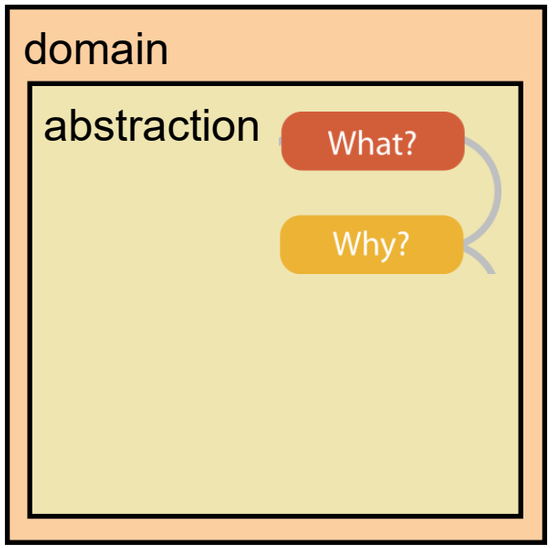


[A Nested Model of Visualization Design and Validation. Munzner. IEEE TVCG 15(6):921-928, 2009 (Proc. InfoVis 2009).]

[A Multi-Level Typology of Abstract Visualization Tasks. Brehmer and Munzner. IEEE TVCG 19(12):2376-2385, 2013 (Proc. InfoVis 2013).]

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- *domain situation*
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- *abstraction*
  - translate from specifics of domain to vocabulary of vis
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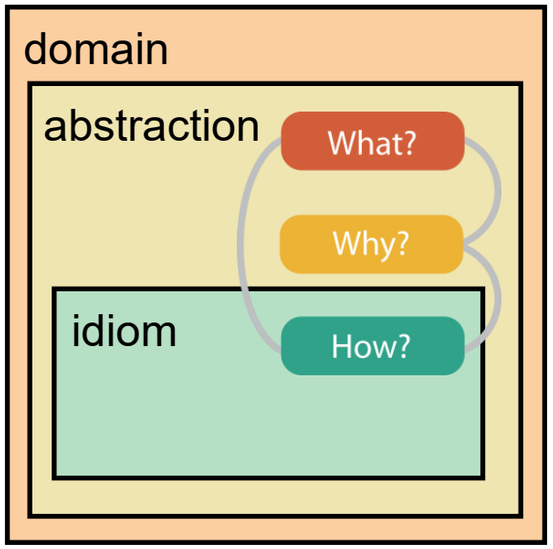


[A Nested Model of Visualization Design and Validation. Munzner. IEEE TVCG 15(6):921-928, 2009 (Proc. InfoVis 2009).]

[A Multi-Level Typology of Abstract Visualization Tasks. Brehmer and Munzner. IEEE TVCG 19(12):2376-2385, 2013 (Proc. InfoVis 2013).]

Analysis framework: Four levels, three questions

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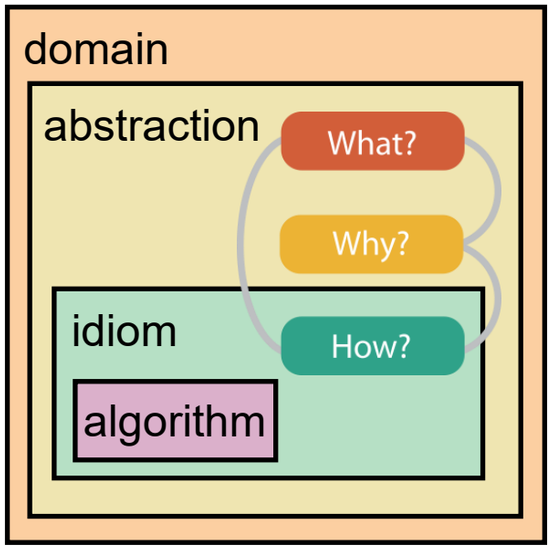


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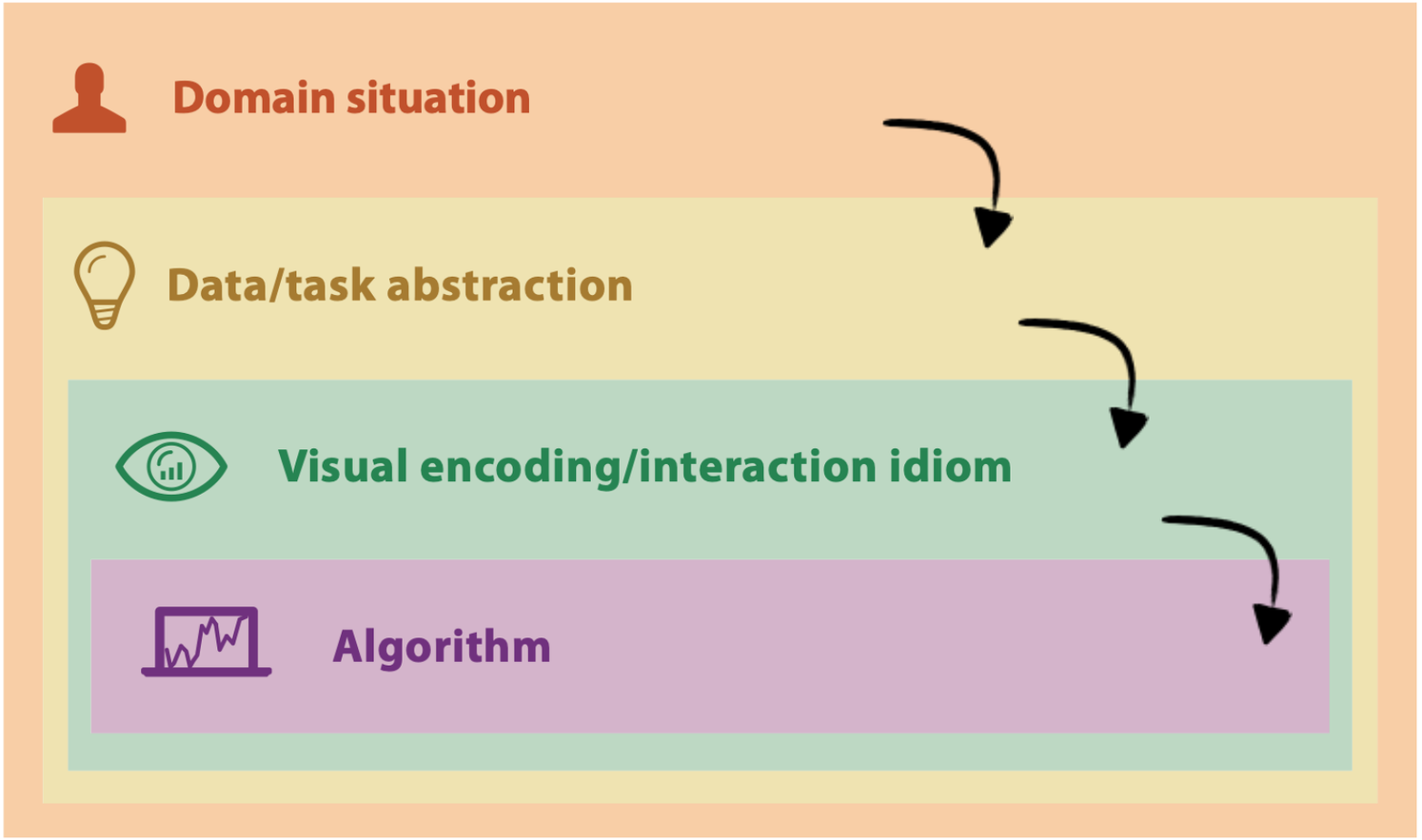


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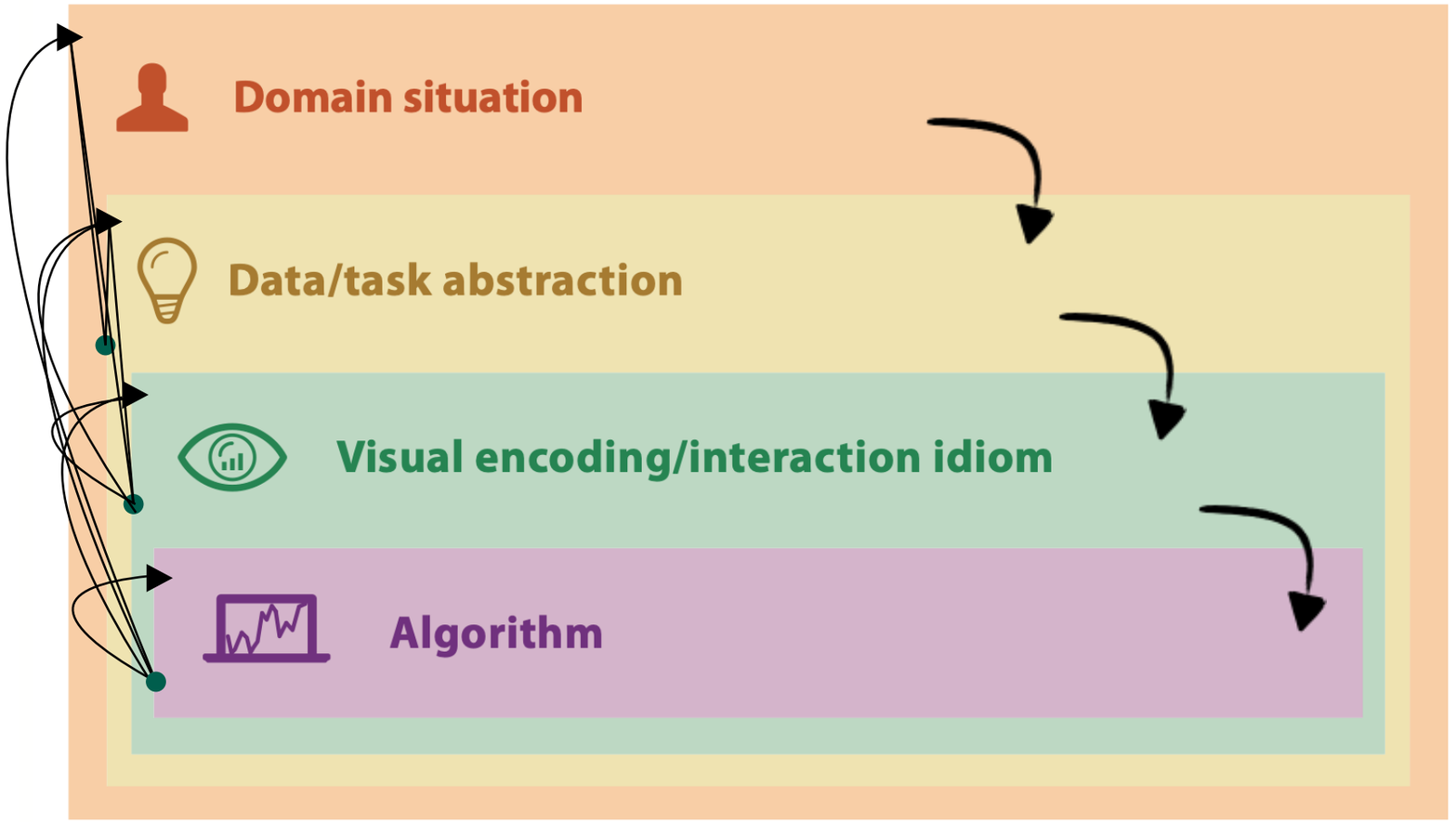
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Nested model

- downstream: cascading effects
- upstream: iterative refinement



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- different ways to get it wrong at each level



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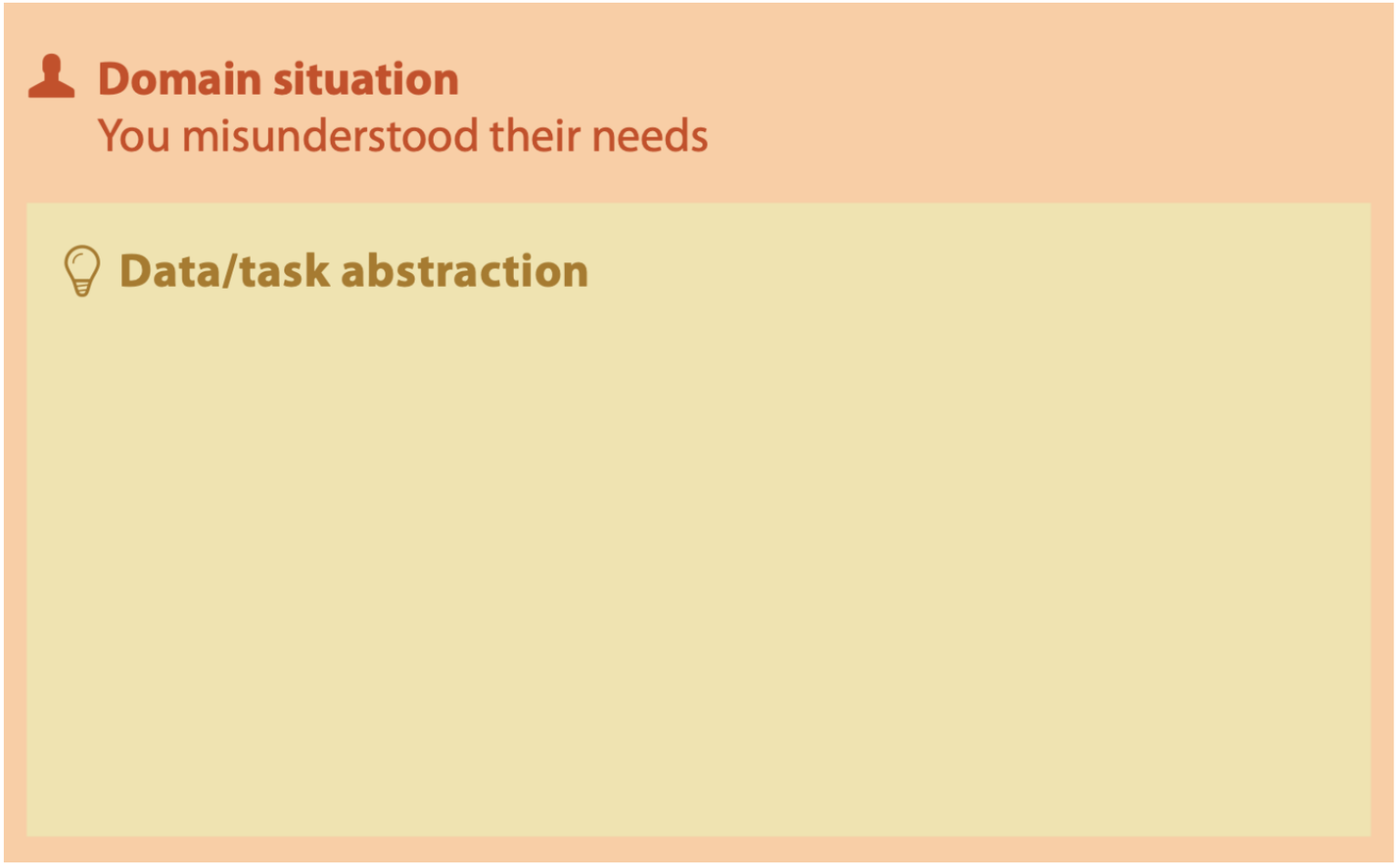
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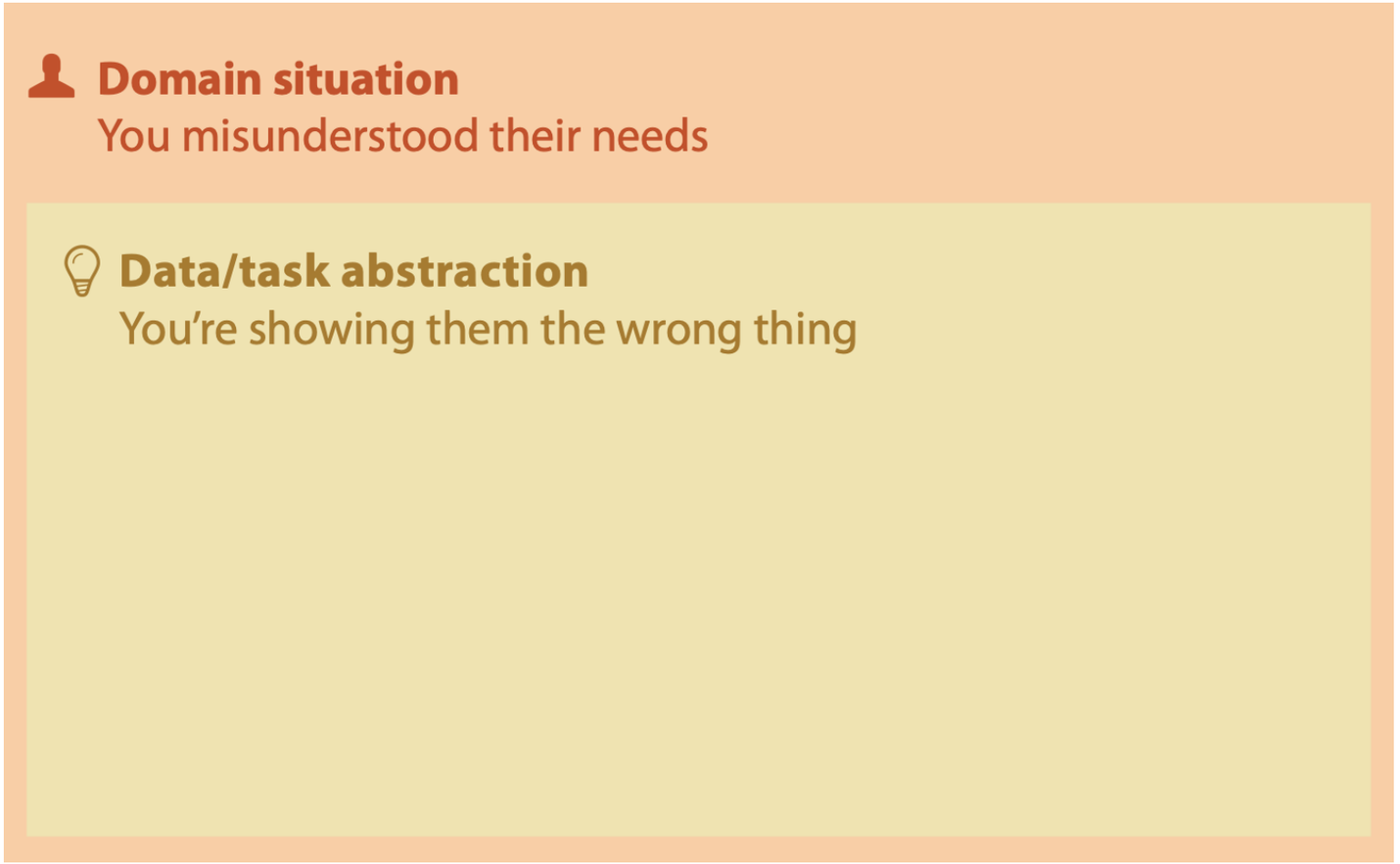
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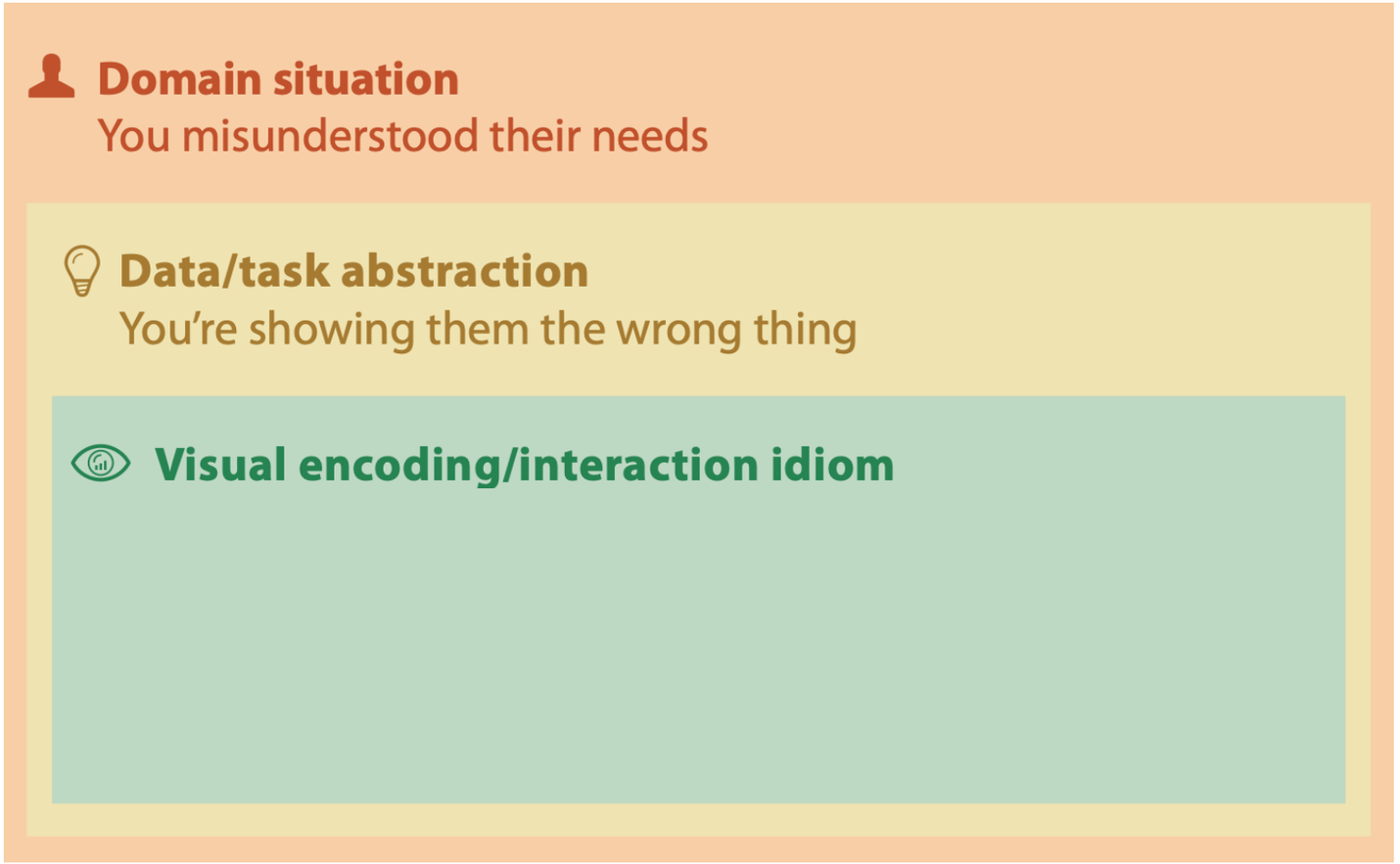
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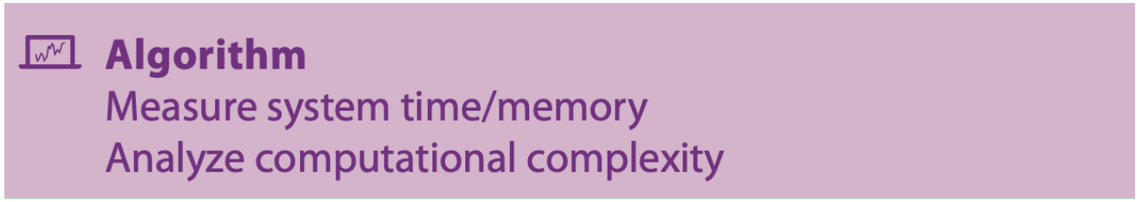
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- solution: use methods from different fields at each level

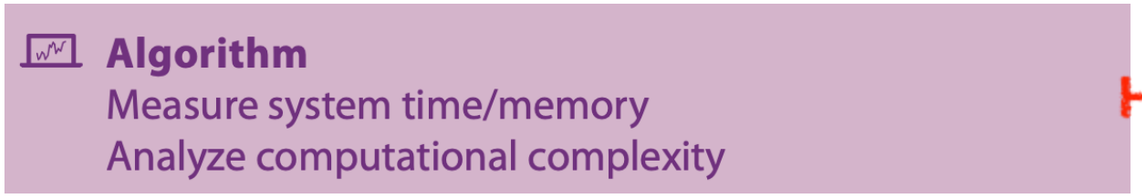


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computer science

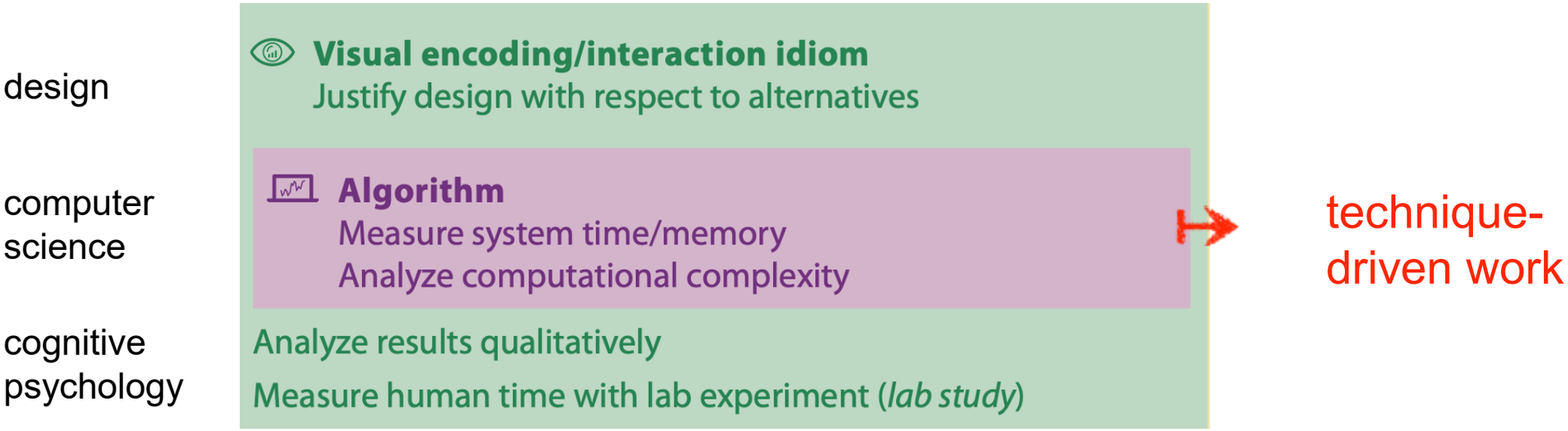


technique-driven work

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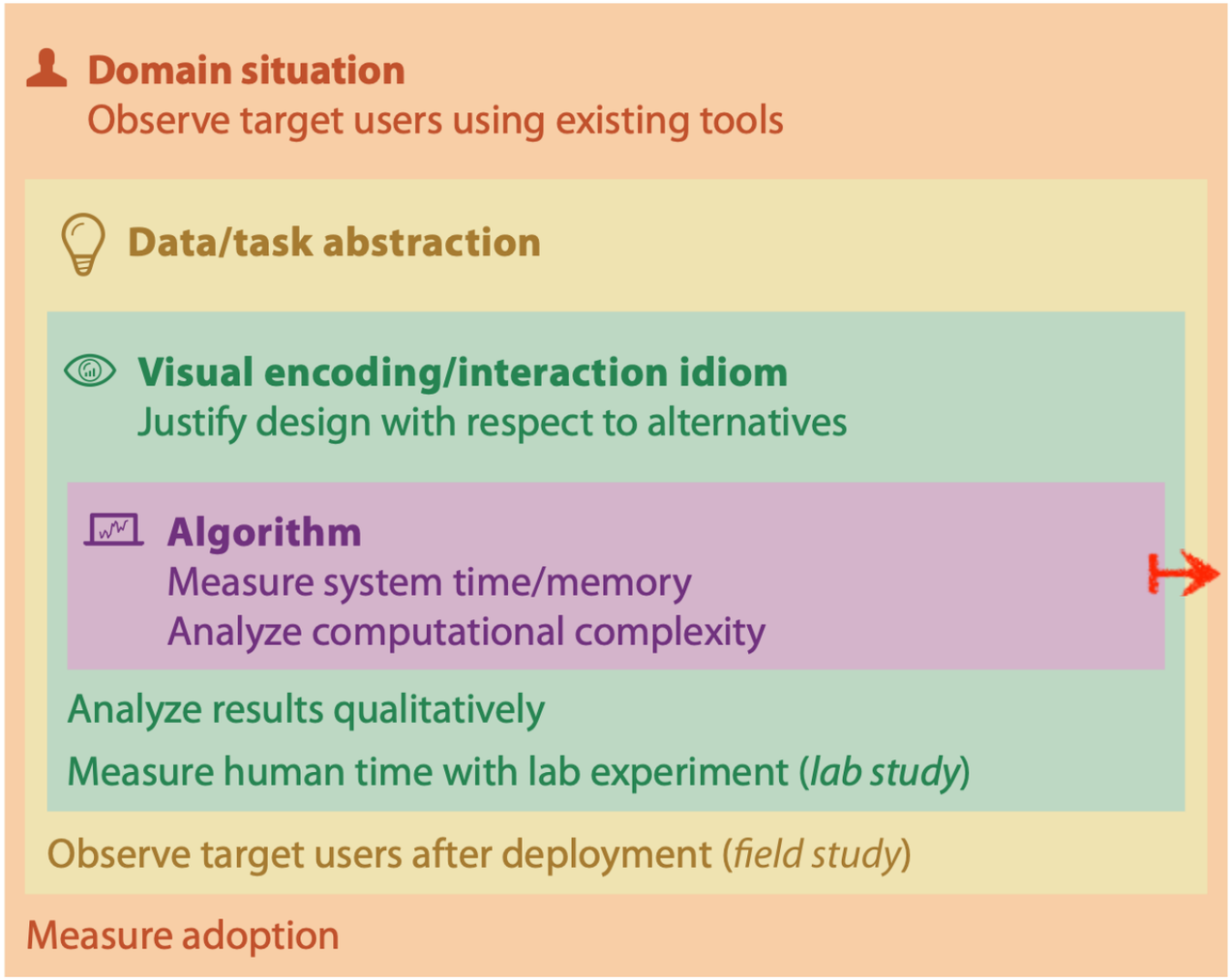
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Agenda

- Task Abstraction (Ch 3 - Tamara's Book)
- Analysis: Nested Model (Ch 4 - Tamara's Book)
- Evaluation (User studies)
- Evaluation (Case studies)

User-centred evaluation

- Assement of a visualization/s to get useful feedback, determine merit, worth, value
- Collecting data on user experiences, task performance etc. identifying strengths and weaknesses in the design, and providing a basis for improvement.
- Investigate/study and analyze how well a visualization supports users' goals and needs

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- **Report** the results considering the tasks, design details, and to validate your proposal

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What is a Case Study (Usage Scenario)

Definition:

- A case study in visualization describes how a visualization system is (or could be) used by real or representative users to accomplish meaningful tasks in a specific context.

Purpose:

- To connect design and use.
- To show how visualization supports real-world reasoning and decision-making.
- To ground evaluation in authentic tasks and contexts.

Example:

- “A city planner uses a traffic visualization to identify congestion hotspots and plan interventions.”

Why Case Studies Matter in Visualization

Bridge between design and evaluation:

- They show how theoretical design choices perform in practice.

Reveal real-world constraints:

- Highlight limitations, data issues, or user needs that lab tests miss.

Communicate value to others:

- Case studies demonstrate impact. How visualizations support discovery or decision-making.

Provide transferable lessons:

- Other designers can learn from what worked (or failed).

Common Mistakes of Case Study

Too generic:

- *“The user explores the data.”*

Too idealized:

- *“The visualization reveals everything instantly.”*

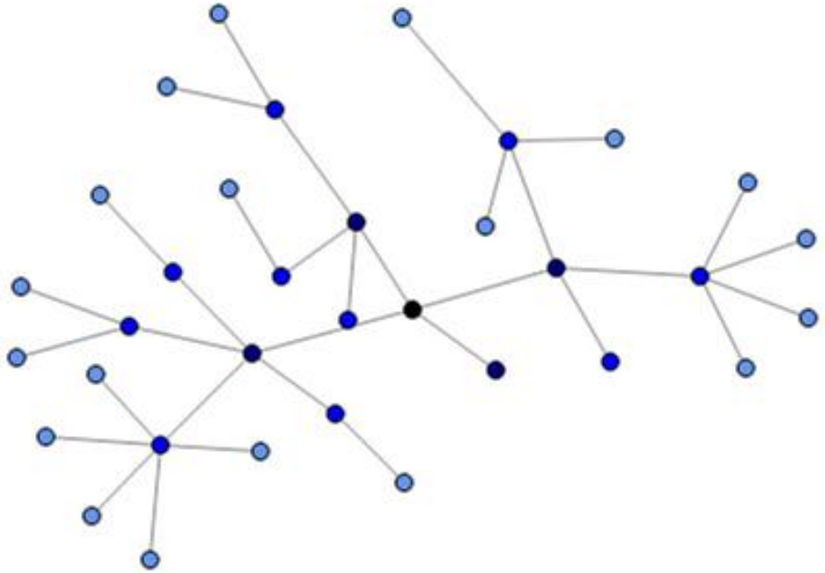
Missing context:

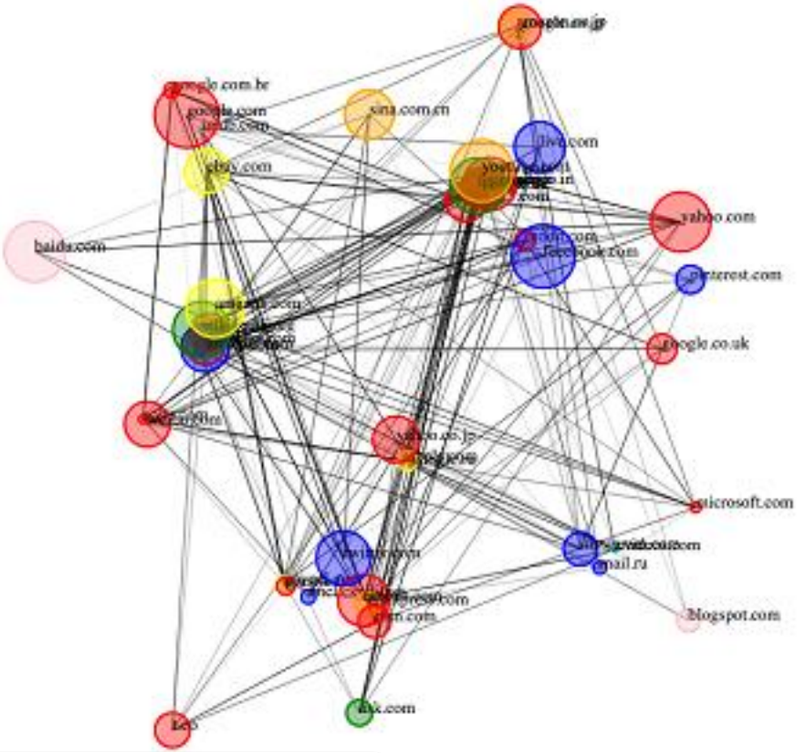
- *Who’s using it, when, or why?*

Overly technical:

- *Focuses on algorithms, not use.*

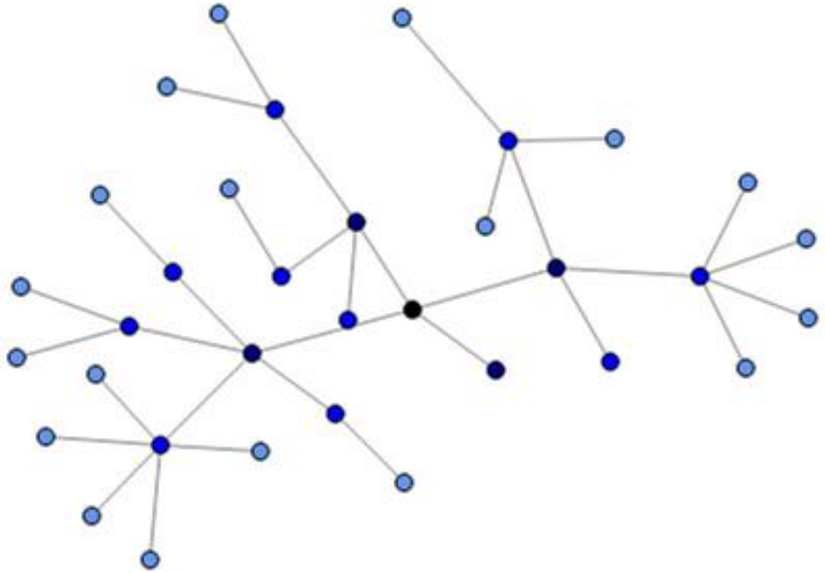
Quantitative evaluation

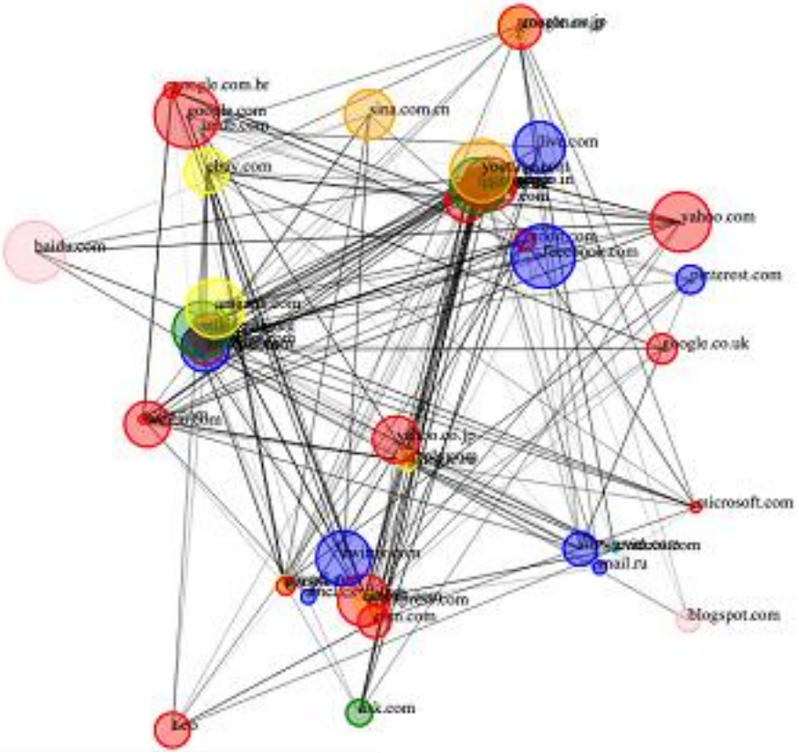




Which one is better?

Quantitative evaluation





How to evaluate the quality of these visualizations?

Further Reading

- **[Mun]** T. Munzner. Visualization Analysis and Design. 1st Edition, A K Peters/CRC Press, ISBN 978-1466508910, 2014.
  - Chapter 3: Why: Task Abstraction
  - Chapter 4: Analysis: Four levels for Validation

